

Course: Algorithmic Trading and Portfolio Management

Machine Learning–Based Trading Strategies for Metal Futures

1. Introduction

This project studies whether machine learning models can be used to improve trading decisions in metal futures markets, specifically COMEX Gold, Silver, and Copper. The focus is on short-term forecasting of next-day price direction and on translating these forecasts into a simple and realistic trading strategy.

Rather than attempting to predict exact prices, the models are designed to answer a more practical question: whether tomorrow's return is likely to be positive or negative. Based on this prediction, a long-or-flat strategy is applied, where the trader either holds a long position or stays in cash. The goal is to evaluate whether such models can provide an edge over a passive buy-and-hold benchmark once risk and drawdowns are taken into account.

The dataset spans approximately fifteen years, from December 2010 to December 2025. This period includes multiple market regimes, such as the post-global financial crisis recovery, commodity downturns, the COVID-19 shock, post-pandemic inflation, monetary tightening, and recent geopolitical disruptions. Using a longer sample allows the models to be tested across diverse economic environments rather than a single unusual regime.

2. Data Description

The analysis uses daily futures data for Gold, Silver, and Copper. Each dataset includes the following variables:

- Date
- Open price
- High price
- Low price
- Close price
- Trading volume
- Daily percentage change

The data was cleaned to remove missing values, converted to numeric format where necessary, and sorted chronologically. No forward-looking information was used at any stage of the analysis.

Gold data range: 2010-12-01 00:00:00 to 2025-12-01 00:00:00 n = 3831
Silver data range: 2010-12-01 00:00:00 to 2025-12-01 00:00:00 n = 3885
Copper data range: 2010-12-01 00:00:00 to 2025-12-01 00:00:00 n = 3870

Gold head:

	date	open	high	low	close	Volume	ChangePct
0	2010-12-01	1387.7	1398.3	1383.0	1388.3	163760.0	0.0016
1	2010-12-02	1387.6	1399.7	1383.8	1389.3	134390.0	0.0007
2	2010-12-03	1387.1	1417.0	1385.3	1406.2	159640.0	0.0122
3	2010-12-06	1415.1	1429.4	1409.8	1416.1	144120.0	0.0070
4	2010-12-07	1426.0	1432.5	1396.7	1409.0	206640.0	-0.0050

Silver head:

	date	open	high	low	close	Volume	ChangePct
0	2010-12-01	28.180	28.880	28.020	28.413	66090.0	0.0071
1	2010-12-02	28.470	29.080	28.330	28.572	64730.0	0.0056
2	2010-12-03	28.625	29.490	28.595	29.271	59920.0	0.0245
3	2010-12-06	29.550	30.355	29.480	29.735	79170.0	0.0159
4	2010-12-07	30.200	30.750	28.485	29.777	104210.0	0.0014

Copper head:

	date	open	high	low	close	Volume	ChangePct
0	2010-12-01	3.8690	3.9445	3.8235	3.9475	90.0	0.0319
1	2010-12-02	3.9450	3.9880	3.9370	3.9790	100.0	0.0080
2	2010-12-03	3.9560	4.0000	3.9550	3.9990	40.0	0.0050
...							
25%		2.757625	2.778625	2.733625	2.758125	240.000000	
50%		3.320750	3.355000	3.291250	3.319000	970.000000	
75%		4.065500	4.102375	4.022000	4.063875	53757.500000	
max		5.844500	5.958500	5.781000	5.819500	253200.000000	

(Figure 2.1: Dataset range and summary statistics for Gold, Silver, and Copper)



(Figure 2.2: Daily closing prices for Gold, Silver, and Copper over the full sample period)

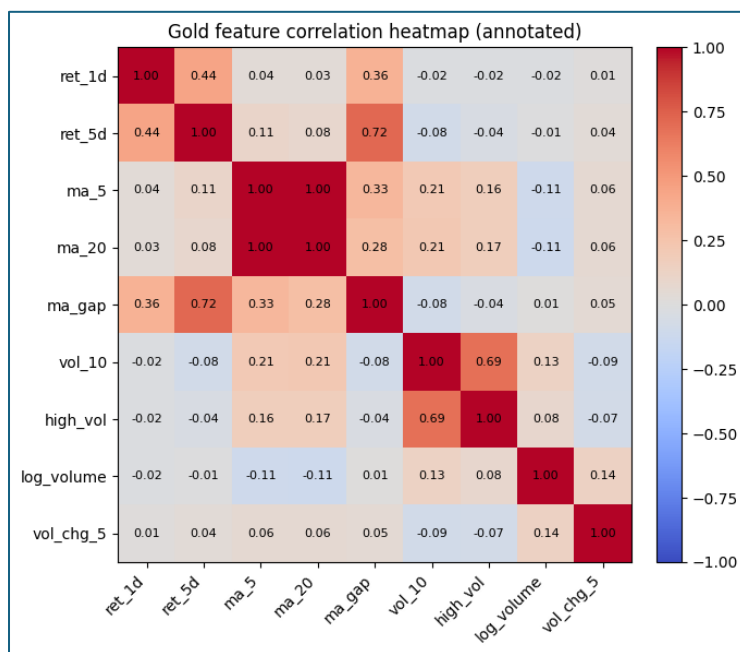
A visual inspection of prices highlights clear differences across assets. Gold exhibits long trending phases, Silver shows sharper spikes and reversals, and Copper displays more cyclical behavior linked to global growth conditions. These differences motivate the use of multiple models and a careful comparison across metals.

3. Feature Engineering

To provide the models with informative inputs beyond raw prices, a set of technical features was constructed. These features are intended to capture short-term momentum, trend strength, volatility, and liquidity conditions.

The final feature set includes:

- *ret_1d*: one-day close-to-close return (short-term momentum)
- *ret_5d*: five-day return (intermediate momentum)
- *ma_5*: five-day moving average (short-term trend)
- *ma_20*: twenty-day moving average (medium-term trend)
- *ma_gap*: difference between the close price and the 20-day moving average (trend deviation)
- *vol_10*: ten-day rolling standard deviation of returns (volatility)
- *high_vol*: binary indicator equal to 1 when *vol_10* is above its median (volatility regime)
- *log_volume*: logarithm of trading volume (scale normalization)
- *vol_chg_5*: five-day percentage change in volume (volume momentum)



(Figure 3.1: Annotated correlation heatmap of features for Gold)

The correlation analysis shows that moving averages are highly correlated with each other, while return, volatility, and volume-based features provide complementary information. This supports the use of tree-based models, which can handle correlated inputs without requiring strong distributional assumptions.

All features are constructed using only past and current information to avoid look-ahead bias. The same feature set is used consistently across all models and assets to ensure a fair comparison.

4. Modeling Approach

The modeling task is framed as a supervised learning problem. For each trading day, the models use the feature vector to predict whether the next day's return will be positive or negative.

Each asset's time series is split chronologically into an 80 percent training period and a 20 percent test period. Models are trained only on the training data and evaluated on the unseen test data. No random shuffling is performed, which preserves the realistic time-series structure of the problem.

Four modeling approaches are considered.

4.1 Random Forest Regressor (RF-sign)

The Random Forest model is trained to predict the next-day return as a continuous value. Instead of using the predicted return directly, its sign is taken to generate a directional signal. A positive prediction implies a long position, while a negative prediction implies staying flat.

This approach allows the model to capture non-linear relationships and interactions between features while keeping the trading rule simple and interpretable.

4.2 K-Nearest Neighbors Classifier (KNN)

The KNN model is trained directly on the binary direction label. Features are standardized so that distance calculations are meaningful. For each new observation, the model looks at the most similar past days and predicts the majority direction.

KNN is included as a transparent baseline. If more complex models cannot outperform such a simple method, their additional complexity may not be justified.

4.3 Ensemble Rule

A simple ensemble combines the Random Forest and KNN signals:

- If both models agree on direction, that signal is used.
- If they disagree, the KNN signal is followed.

This rule is intentionally straightforward. The goal is to improve stability by requiring agreement for strong signals while relying on the more robust model when there is disagreement.

4.4 XGBoost Classifier

XGBoost is used as a more flexible, gradient-boosted tree classifier trained on the binary direction label. To reduce overtrading, a probability threshold is applied: a long position is taken only when the predicted probability of an upward move exceeds 0.55. Otherwise, the strategy remains flat. In practice, this fixed threshold is replaced by an adaptive, rolling quantile threshold to prevent prolonged periods of inactivity, as discussed in Section 5.

5. Trading Strategy and Backtesting

All models are evaluated using the same long-or-flat trading rule:

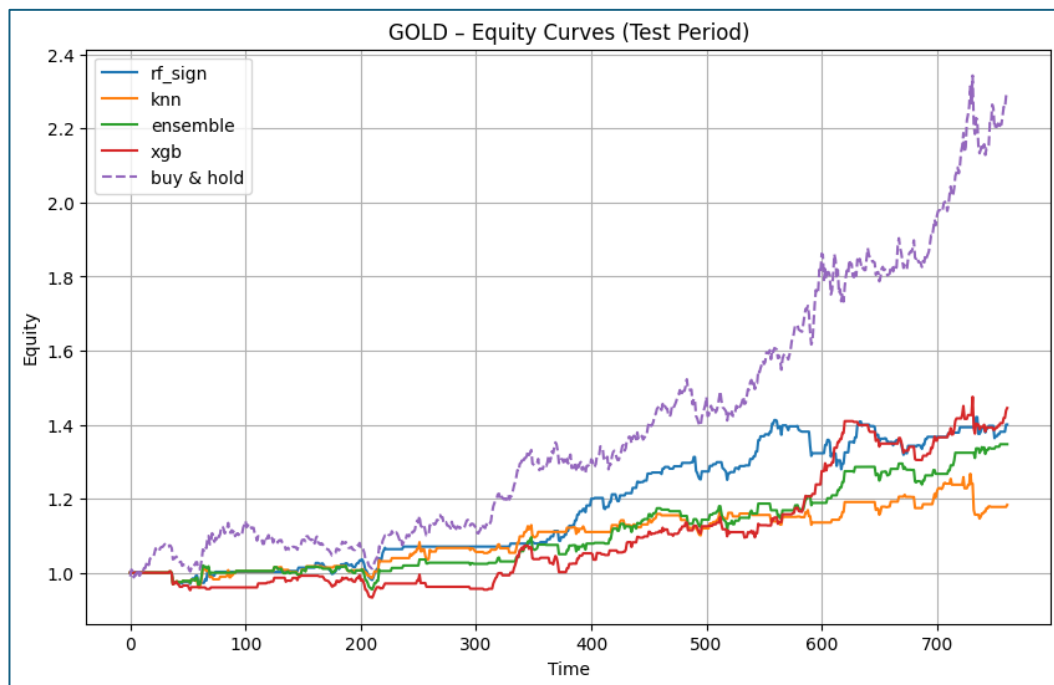
- Go long when the model signals an upward move.
- Stay in cash otherwise.

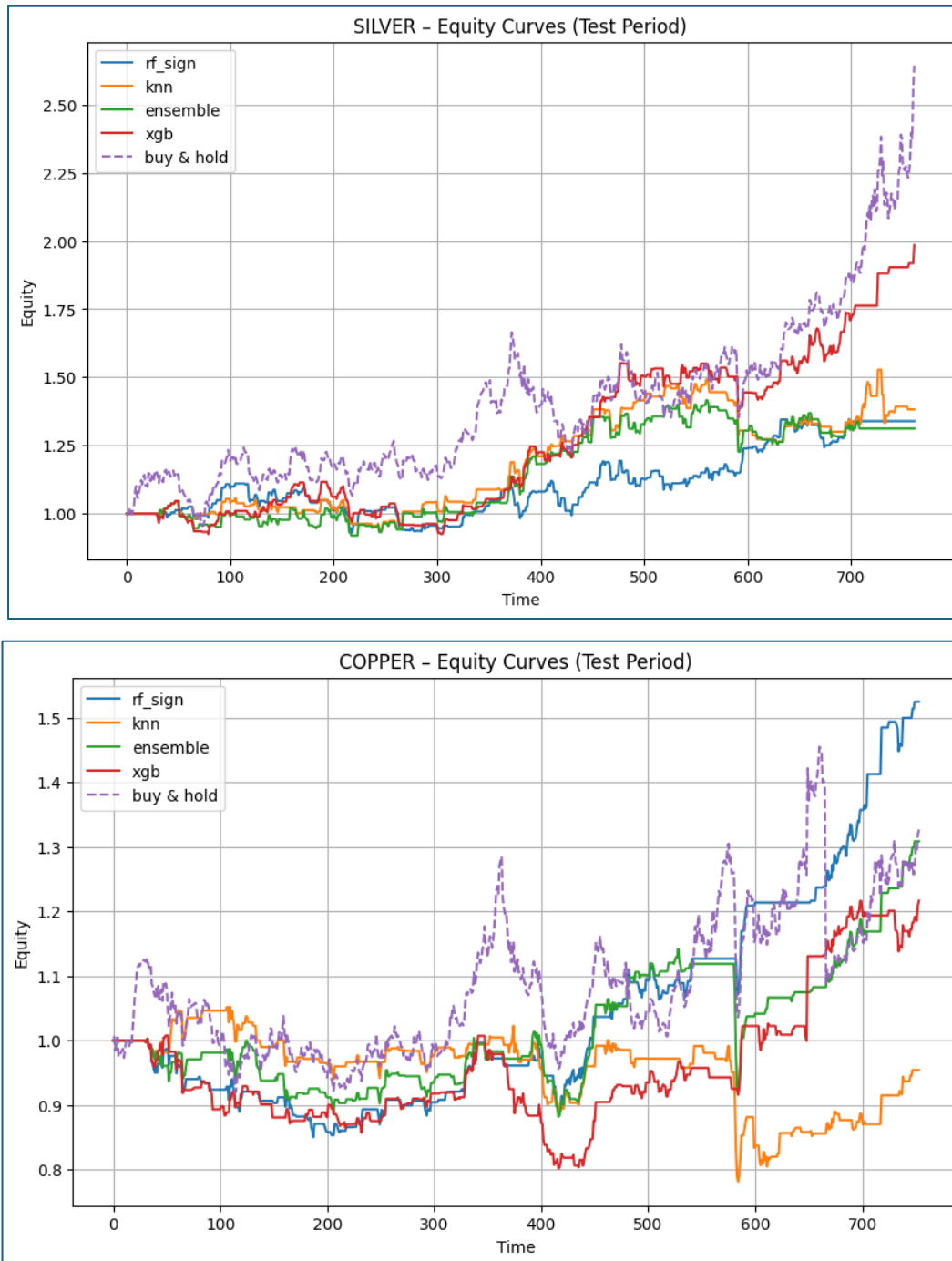
Daily strategy returns are computed by multiplying the trading signal by the realized next-day return. Performance is evaluated using:

- Annualized return
- Annualized volatility
- Sharpe ratio (assuming a zero risk-free rate)

Each strategy is compared against a buy-and-hold benchmark for the same asset.

To address issues of prolonged inactivity, adaptive signal thresholds are used so that models trade when their predictions are strong relative to recent history. Additional diagnostics such as the number of trades, active percentage of days, maximum flat periods, and profit and loss streaks are also reported.





(Figure 5.1: Equity curves for all models and buy-and-hold across Gold, Silver, and Copper)

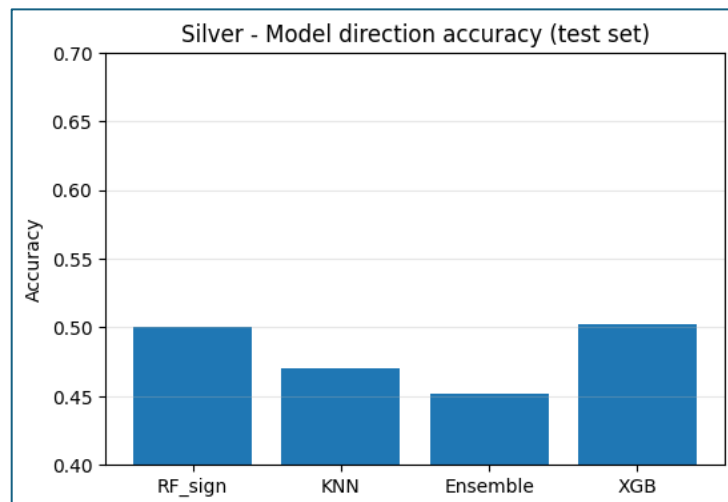
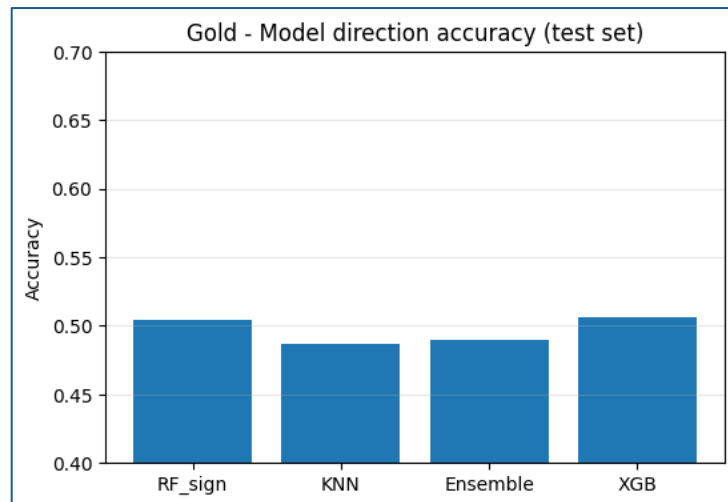
6. Results and Discussion

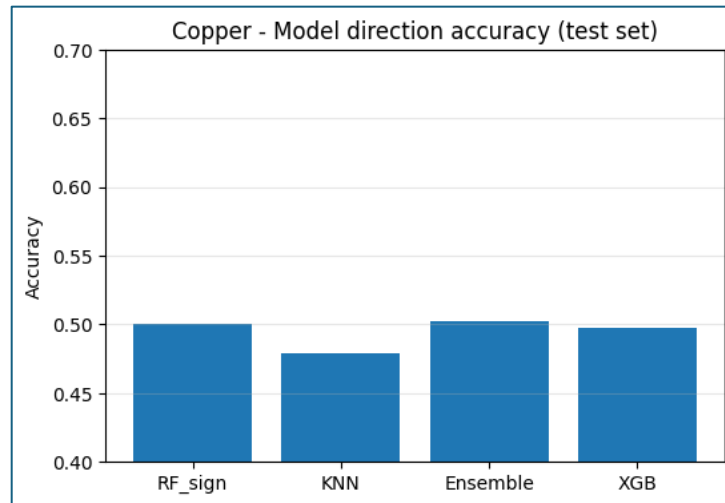
6.1 Directional Accuracy

Across all three metals, directional accuracy remains close to 50 percent, which is consistent with the difficulty of predicting daily price movements in financial markets.

- Gold: Model accuracies cluster around 49–51 percent, with XGBoost and RF-sign performing slightly better than KNN and the ensemble.
- Silver: All models show similar accuracy, again near 50 percent, reflecting Silver’s more erratic price behavior.
- Copper: Accuracy differences are small, with the ensemble marginally outperforming other models.

While these accuracy levels may appear low, even small statistical edges can be meaningful when combined with disciplined trading rules and risk controls.





(Figure 6.1: Model direction accuracy by asset)

6.2 Trading Performance

Trading performance reveals clearer differences than raw accuracy.

- Gold: All model-based strategies deliver positive Sharpe ratios, though buy-and-hold remains strong during the test period. Model-based strategies achieve smoother equity curves by avoiding some drawdowns.
- Silver: Buy-and-hold benefits from large rallies, but the XGBoost-based strategy achieves a competitive Sharpe ratio by selectively entering the market.
- Copper: The Random Forest sign-based strategy performs particularly well on a risk-adjusted basis, suggesting that simpler models may capture cyclical behavior more effectively.

Overall, strategies with modest accuracy but controlled exposure tend to perform better on a risk-adjusted basis than more aggressive approaches.

Performance metrics (test period):							
	asset	model	ann_ret	ann_vol	sharpe	bh_ann_ret	bh_sharpe \
0	gold	rf_sign	0.118	0.090	1.308	0.317	1.924
1	gold	knn	0.057	0.090	0.633	0.317	1.924
2	gold	ensemble	0.104	0.084	1.229	0.317	1.924
3	gold	xgb	0.129	0.098	1.319	0.317	1.924
4	silver	rf_sign	0.101	0.156	0.651	0.379	1.299
5	silver	knn	0.113	0.169	0.669	0.379	1.299
6	silver	ensemble	0.094	0.167	0.563	0.379	1.299
7	silver	xgb	0.254	0.172	1.476	0.379	1.299
8	copper	rf_sign	0.151	0.125	1.207	0.099	0.368
9	copper	knn	-0.016	0.138	-0.113	0.099	0.368
10	copper	ensemble	0.094	0.135	0.696	0.099	0.368
11	copper	xgb	0.068	0.146	0.463	0.099	0.368
	num_trades	win_rate	max_flat_days	active_pct	max_loss_streak \		
0	232	0.579	92	0.308	9		
1	217	0.555	34	0.286	4		
2	222	0.553	35	0.323	9		
3	253	0.571	40	0.358	8		
4	322	0.555	58	0.360	7		
5	258	0.514	30	0.319	4		
6	280	0.488	52	0.374	4		
7	279	0.554	30	0.391	5		
...							
8	4						
9	3						
10	4						
11	5						

(Insert Figure 6.2: Performance metrics across assets and models (test period))

6.3 Feature Importance and Model Comparison

Feature importance analysis shows consistent patterns across assets:

- Short-term returns, moving averages, and volatility measures are the most influential predictors.
- Volume-based features play a larger role in Copper, consistent with its sensitivity to industrial demand.
- The `high_vol` indicator contributes relatively little once continuous volatility measures are included.

In terms of model comparison:

- Simpler models such as KNN and RF-sign often generalize better across assets.
- The ensemble provides stable performance without relying on complex tuning.
- XGBoost performs well in some cases but does not consistently dominate, highlighting the risk of overfitting in noisy financial data.

7. Key Insights

- Small predictive edges can translate into meaningful risk-adjusted returns when paired with conservative trading rules.
- Model simplicity often improves robustness, especially across changing market regimes.
- Different metals respond to different dynamics: Gold trends more strongly, Silver is volatile and difficult to time, and Copper exhibits cyclical behavior linked to macroeconomic conditions.
- Visual diagnostics and trading activity metrics are essential for understanding how and when models add value.

8. Limitations

- Despite the extended sample, results may still depend on the specific market regimes observed.
- Transaction costs, slippage, and margin requirements are not included and would reduce real-world returns.
- Only long-or-flat strategies are considered; short-selling and dynamic position sizing are excluded.
- Hyperparameter tuning is limited in order to reduce overfitting, but this may leave some performance unrealized.

9. Conclusion

This project demonstrates that machine learning models, even relatively simple ones, can provide useful timing signals in metal futures markets when applied carefully. While directional prediction accuracy remains modest, disciplined long-or-flat strategies can improve risk-adjusted performance and reduce drawdowns compared to passive benchmarks.

The results reinforce a key lesson from quantitative finance: complexity does not guarantee superior performance. Transparent models paired with intuitive trading rules often generalize better than highly flexible algorithms. Overall, the project provides a solid foundation for further experimentation with richer features, alternative horizons, and more advanced portfolio construction techniques.

10. References

COMEX historical futures data for Gold, Silver, and Copper, sourced from [www.Investing.com](http://www.investing.com)

Python libraries used:

- pandas
- NumPy
- scikit-learn
- XGBoost
- matplotlib

Additional references:

- scikit-learn documentation
- XGBoost documentation
- Investopedia (Sharpe ratio, backtesting concepts)